Does a Protestant Work Ethic Exist?

A Micro-Level Analysis of Religious Beliefs and Economic Outcomes

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Outline

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Protestantism in History

In 1517, Martin Luther posted his 95 Theses to the door of the Castle Church in Wittenberg, challenging the Catholic Church's practice of indulgences. This act marked the beginning of the Protestant Reformation — one of the most significant religious movements in Western history.

Protestantism spread rapidly across Europe and, in subsequent centuries, worldwide. Today, approximately 800 million people identify as Protestant, making it one of the most influential religious traditions in the world.

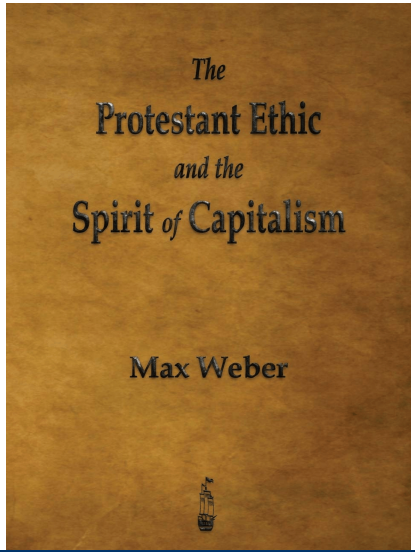


Martin Luther (1483–1546)

Weber's Theory

Since Weber (1905), scholars have hypothesized that religious identities affect individuals' economic outcomes. In *The Protestant Ethic and the Spirit of Capitalism*, Weber argued that Protestant doctrine fostered values of hard work, thrift, and discipline that were conducive to the development of capitalism.

Weber's central claim is that the Protestant Reformation altered the work ethic in Northern Europe, encouraging individuals to produce more, accumulate wealth, and devote themselves to their occupations as a way of glorifying God. Scholars have continued to debate these correlations between religion and economic outcomes ever since.



Literature Review

The relationship between religion and economic outcomes has been examined from multiple perspectives:

- **Cross-country evidence:** Much of the existing evidence is based on cross-country studies in which the impact of religion is confounded by differences in other institutional factors (Barro & McCleary, 2003).
- **Micro-level analysis:** Individual-level studies offer more precise identification by controlling for individual characteristics, separating religious effects from broader institutional ones.
- **Causal identification:** Alternative strategies include laboratory experiments varying religious identity salience (Benjamin, Choi & Fisher, 2016) and instrumental variable approaches exploiting exogenous geographic variation.

Theoretical Mechanism

Religion influences economic outcomes through multiple transmission channels, as outlined by Öhlmann (2021):



Source: Adapted from Öhlmann (2021) and Guiso, Sapienza & Zingales (2003)

Research Objective and Identification Challenge

Objective: This study aims to estimate the causal effect of Protestant affiliation on work ethic, one of the key individual economic attitudes theorized by Weber (1905), using micro-level data from 47 European countries.

Identification Challenge: Measuring the causal effect of religion has been hampered by the difficulty of identifying exogenous variation in religious identities (Benjamin, Choi & Fisher, 2016).

- Religious affiliation is likely correlated with unobserved background factors.
- The observed correlation between work ethic and Protestant affiliation could reflect selection of individuals with stronger work ethic into Protestantism.

Data — Integrated Values Survey

Dataset

- Integrated Values Survey (IVS)
- EVS + WVS combined
- Waves 3–5: 1999, 2008, 2017
- 47 European countries
- 163,242 observations
- Regression sample: ∼131,500

Instrument

- Log distance from country capital to Wittenberg, Germany
- Following Becker & Woessmann (2009)
- First stage F-stat: 1,526

Key Variables

- *Work ethic index:* Mean of 4 standardized items
- *Protestant:* Dummy = 1 if Protestant denomination
- *Attendance:* Current religious service frequency (1–8)
- *Childhood attendance:* Religious practice at age 12 (1–8)
- *Belief index:* Mean of beliefs in God, afterlife, hell, heaven (Davis & Rodriguez 2024)

Hypotheses

H1 — Protestant Work Ethic

Protestants exhibit a stronger work ethic than other religious groups, consistent with Weber's (1905) thesis.

$$WorkEthic_{ict} = \alpha + \beta_1 Protestant_{ict} + \sum_k \gamma_k Denom^k_{ict} + \delta Attend_{ict} + \mathbf{X}'_{ict}\phi + \mu_c + \lambda_t + \varepsilon_{ict}$$

Expectation: $\hat{\beta}_1 > 0$ and statistically significant.

i = individual · c = country · t = wave · $\sum_k \gamma_k Denom^k$ = denomination dummies · μ_c = country FE · λ_t = wave FE · $\mathbf{X}'$ = gender, age, education, income

Hypotheses

H2 — Beliefs vs Affiliation

Religious beliefs shape work ethic more than denominational affiliation, following Davis & Rodriguez (2024).

$$\begin{aligned} \mathit{WorkEthic}_{ict} = \alpha + \beta_1 \mathit{Protestant}_{ict} + \sum_k \gamma_k \mathit{Denom}^k_{ict} + \delta_1 \mathit{Attend}_{ict} + \delta_2 \mathit{ChildAttend}_{ict} \\ + \beta_2 \mathit{BeliefIndex}_{ict} + \mathbf{X}'_{ict}\phi + \mu_c + \lambda_t + \varepsilon_{ict} \end{aligned}$$

Expectation: $\hat{\beta}_1$ shrinks when $\mathit{BeliefIndex}_{ict}$ is added, while $\hat{\beta}_2 > 0$ and statistically significant.

β_2 = effect of belief index · δ_1, δ_2 = attendance effects · μ_c = country FE · λ_t = wave FE

Empirical Strategy

OLS — Progressive Specification:

- **H1:** Add attendance and childhood attendance progressively
- **H2:** Add belief index — test if Protestant effect shrinks
- Country + wave fixed effects throughout
- SE clustered at country level

IV — 2SLS with Distance to Wittenberg:

First stage:

$$Protestant_{ict} = \pi_1 \ln(Distance_c) + \delta_1 Attend_{ict} + \delta_2 ChildAttend_{ict} + \mathbf{X}'_{ict}\phi + \lambda_t + \nu_{ict}$$

Second stage:

$$WorkEthic_{ict} = \alpha + \beta_1^{IV} \widehat{Protestant}_{ict} + \delta_1 Attend_{ict} + \delta_2 ChildAttend_{ict} + \mathbf{X}'_{ict}\phi + \lambda_t + \varepsilon_{ict}$$

$\mathbf{X}_{ict}$: gender, age, age^2, education, income · μ_c: country FE (OLS only) · λ_t: wave FE

Methodological Approach

- **Country FE:** Absorb time-invariant country characteristics, addressing the limitation of cross-country studies where institutional differences confound religious effects.
- **Wave FE:** Control for common macroeconomic shocks affecting all countries in a given survey wave.
- **Belief index:** Disentangles the effect of religious affiliation from religious beliefs (Davis & Rodriguez, 2024).

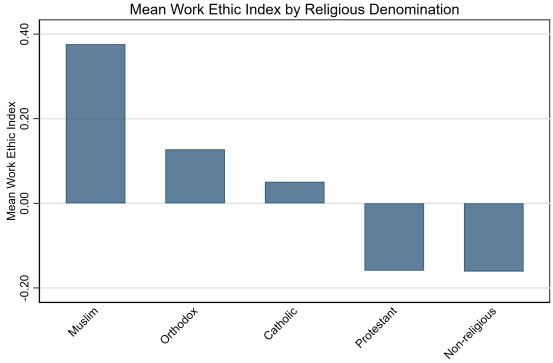
Why Wittenberg?

A valid instrument must satisfy two conditions:

- **Relevance:** The instrument must be correlated with the endogenous variable — Protestant affiliation.
- **Exogeneity:** The instrument must affect work ethic *only through* Protestant affiliation and not through any other channel.

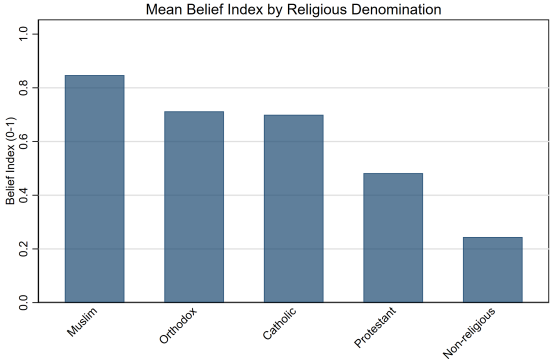
Descriptive Statistics

Figure 1: Work Ethic Index by Denomination



Note: EVS waves 3-5, European countries.
Work ethic index standardized with mean 0.

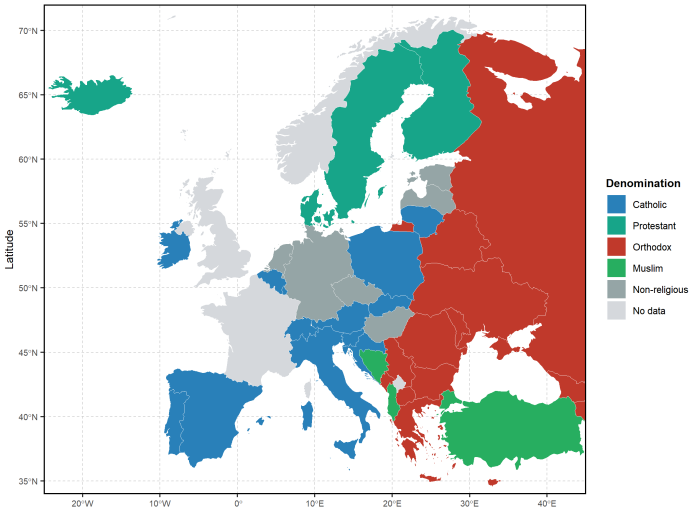
Figure 2: Belief Index by Denomination



Note: Belief index = mean of beliefs in God, afterlife, hell and heaven (Davis & Rodriguez 2024).

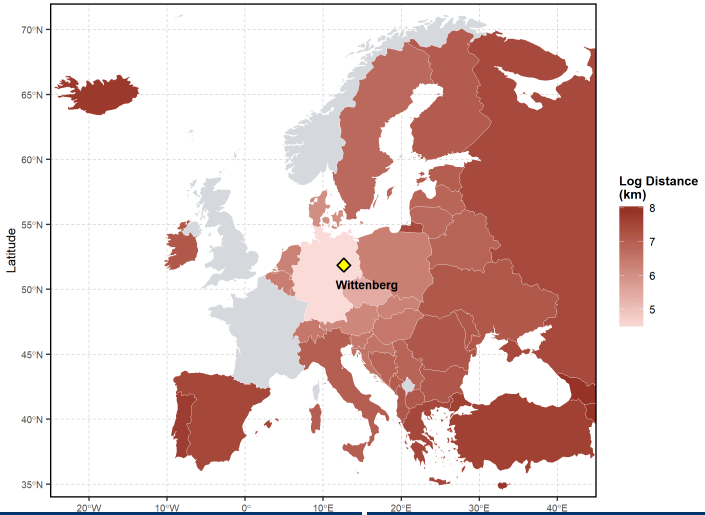
Descriptive Statistics

Map 2: Dominant Religious Denomination by Country



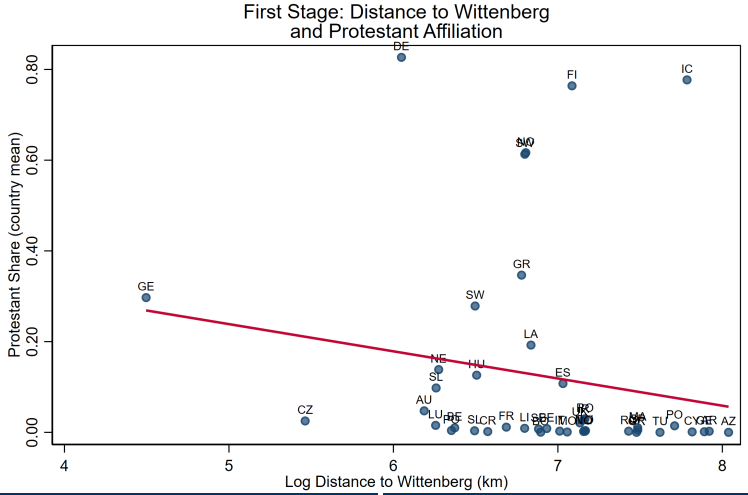
IV Strategy — Distance to Wittenberg

Map 2: Log Distance to Wittenberg by Country



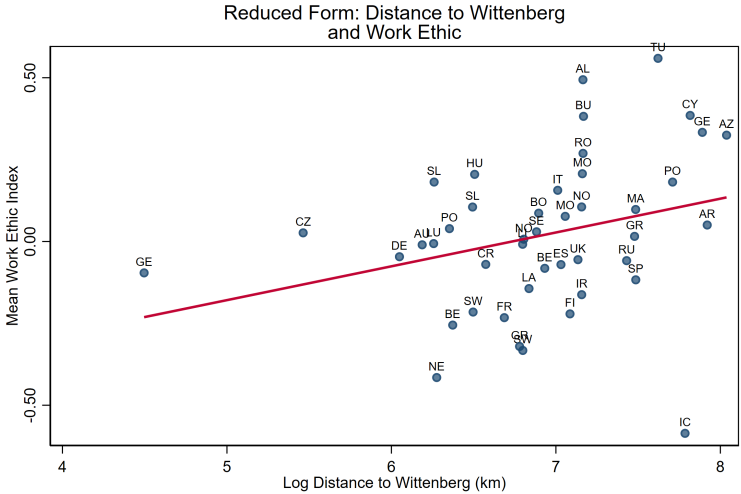
First Stage

Figure 4: First Stage — Distance to Wittenberg and Protestant Share



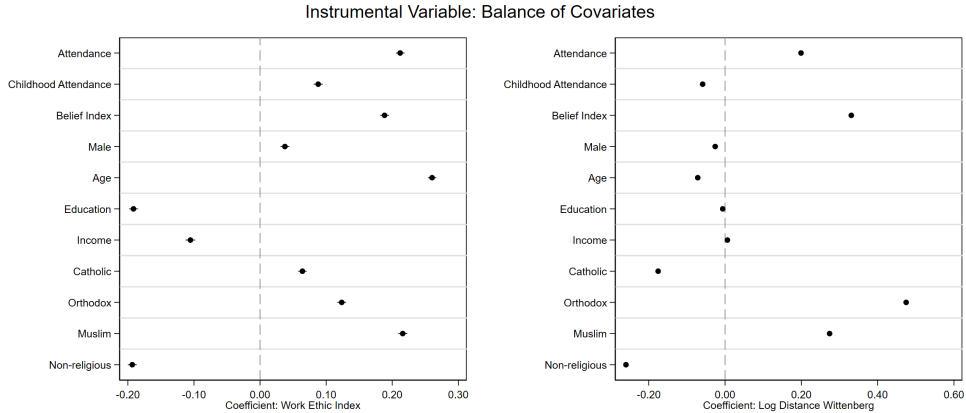
Reduced Form

Figure 5: Reduced Form — Distance to Wittenberg and Work Ethic



Balance of Covariates

Figure 6: Balance of Covariates



Note: All variables standardized to mean 0 SD 1.
Dots show OLS coefficient with 95% confidence intervals.
Left panel: each covariate regressed on work ethic index.
Right panel: each covariate regressed on log distance to Wittenberg.
Dashed line at zero.

Table 1: Protestant Work Ethic — OLS Results

Table 1: Protestant Work Ethic and Religious Attendance

	(1)	(2)	(3)
Key Variables:			
Protestant	0.084***	0.054***	0.055***
	(0.018)	(0.018)	(0.018)
Attendance		0.016***	0.015***
		(0.002)	(0.002)
Childhood Attendance			0.004**
			(0.002)
Individual controls	✓	✓	✓
Denomination controls	✓	✓	✓
Country FE	✓	✓	✓
Wave FE	✓	✓	✓
Observations	136,149	135,101	130,662
R-squared	0.169	0.171	0.172

Table 2: Beliefs vs Affiliation — OLS Results

Table 2: Religious Beliefs and Work Ethic

	(1)	(2)	(3)
Key Variables:			
Protestant	0.055***	0.045**	
	(0.018)	(0.018)	
Attendance	0.015***	0.012***	0.015***
	(0.002)	(0.002)	(0.002)
Childhood Attendance	0.004**	0.004**	0.006***
	(0.002)	(0.002)	(0.002)
Belief Index		0.044***	0.068***
		(0.012)	(0.013)
Individual controls	✓	✓	✓
Denomination controls	✓	✓	
Country FE	✓	✓	✓
Wave FE	✓	✓	✓
Observations	130,662	127,285	127,655
R-squared	0.172	0.174	0.172

*** p<0.01, ** p<0.05, * p<0.10

Table 3: IV Results — Distance to Wittenberg

Table 3: IV Estimates — Distance to Wittenberg

	(1) First Stage	(2) 2SLS	(3) 2SLS
Log Distance Wittenberg	−0.062***		
	(0.002)		
Protestant		0.140	0.253
		(0.401)	(0.232)
Attendance	−0.003***	0.022*	0.018***
	(0.000)	(0.012)	(0.005)
Childhood Attendance	−0.006***	−0.010***	−0.011***
	(0.000)	(0.004)	(0.002)
Denomination controls		✓	✓
Wave FE	✓	✓	✓
Belief Index			✓
Observations	129,498	127,896	124,560
R-squared	0.030	0.105	0.096
First stage F-stat	1,525.87		

*** p<0.01, ** p<0.05, * p<0.10

Robust SE. No country FE — distance is country-level variable.

Conclusions

- **H1 supported:** OLS estimates show that Protestants exhibit significantly stronger work ethic than other religious groups ($\hat{\beta} = 0.054^{***}$), consistent with Weber's (1905) thesis.
- **H2 partially supported:** Adding the belief index reduces the Protestant coefficient by 17% (0.054 $\rightarrow$ 0.045**), suggesting that both affiliation and beliefs independently shape work ethic, consistent with Davis & Rodriguez (2024).
- **IV results:** The 2SLS estimate of the Protestant coefficient is positive but insignificant (0.140, p=0.727).

Limitations and Future Research

- **Instrument for belief index:** While we instrument for Protestant affiliation, the belief index may itself be endogenous. A separate instrument for religious beliefs would be required to fully identify the causal effect of beliefs on work ethic.
- **Geographic scope:** The original intention was to extend the analysis worldwide. However, potential instruments such as natural disaster exposure affect employment directly, violating the exclusion restriction for a work ethic outcome.
- **Country-level instrument:** Using distance to Wittenberg at the country level prevents the inclusion of country fixed effects in the IV specification, restricting identification to between-country variation only.
- **Future research:** Regional-level instruments, panel data on religious switching, and alternative outcome variables such as labor market participation could extend this analysis.

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